Lucas Zhao

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Education:

Computing: Graphics & Visualization Master's Degree, University of Utah, Salt Lake City, UT
Aug 2025 - present

Computer Science Bachelor's Degree, Rutgers University, Piscataway, NJ 2021 – May 2025
GPA: 3.626/4.0

Work Experience:

Data & Application Integration Intern at Application Consulting Group 01/2026 – 01/2026

- Managed and optimized TM1 metadata (dimensions, hierarchies) through TurboIntegrator scripts, ensuring data integrity and alignment with evolving business structures.
- Developed and maintained ETL processes to manipulate cube data and dimensions, enabling accurate and timely monthly financial consolidation and reporting.
- Leveraged TurboIntegrator to build automated data flows, transforming raw source data into actionable management reports within the TM1 cube environment.

Digital PV/CMO Intern at Sanofi, Bridgewater, NJ 06/2024 – 08/2024

- Generated detailed reports on Pharmacovigilance and accessibility, contributing to regulatory compliance and improved decision-making.
- Developed and implemented a Python script to automate data extraction from JSON files into Excel, enhancing workflow productivity.
- Leveraged expertise in data entry to ensure accuracy and efficiency in handling large datasets.

Enterprise Planning Intern at AP&G Co., Inc., Bayonne NJ 06/2023 – 08/2023

- Collaborated cross-functionally with team members across diverse departments to formulate and enhance manufacturing process flows, meticulously documenting the refined processes using Visio.
- Devised strategic methods for efficiently gathering, sorting, and documenting product numbers, optimizing data management protocols.

Projects:

- Engineered a real-time infinite terrain renderer in C++ and OpenGL 4.1 using GPU Geometry Clipmaps, implementing procedural ridge fBm noise synthesis with domain warping and toroidal texture addressing to stream an endless, non-repeating mountain landscape at 120+ FPS
- Developed MATLAB defect detection system utilizing edge detection, RGB analysis, and k-means clustering to identify manufacturing flaws in production line imagery
- Designed and implemented property-based test suites in TypeScript and Python to identify engine-level bugs in Pokémon battle mechanics, using multiple testing strategies to isolate speed calculation overflow errors and AI script logic faults across 50,000+ randomised trials
- Developed a Python-based analytics tool using SQLite and statistical models to analyze and simulate gacha system mechanics, predicting revenue performance and identifying key factors influencing character revenue in Honkai Star Rail.

Technical Skills:

- C++, C, Java, Python, Matlab, SQLite, x86 Assembly
- Blender/3D modeling, Unreal Engine 5, Maya, Houdini
- OpenGL